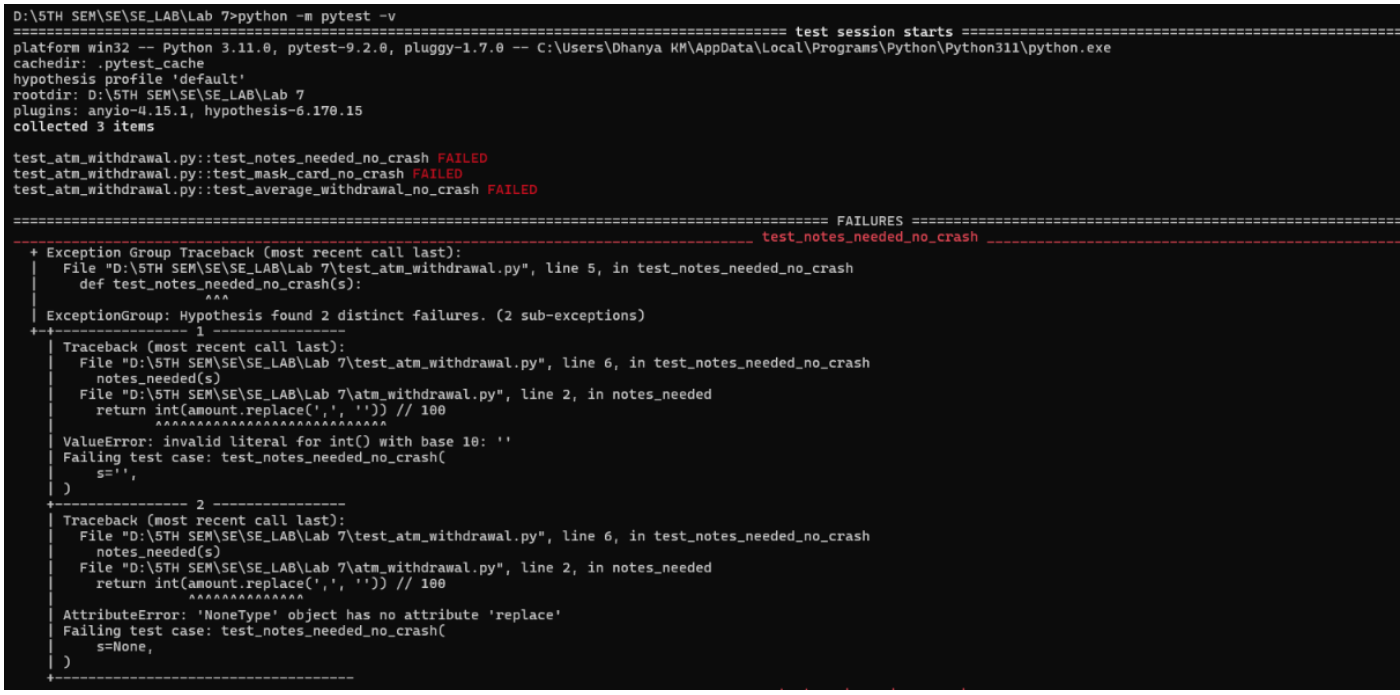
Name: Dhanya K M

SRN: PES1UG24CS567

BEFORE FIXING THE BUG



```
D:\5TH SEM\SE\SE_LAB\Lab 7>python -m pytest -v
============================================================================== test session starts ==============================================================================
platform win32 -- Python 3.11.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Dhanya KM\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: D:\5TH SEM\SE\SE_LAB\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 3 items

test_atm_withdrawal.py::test_notes_needed_no_crash FAILED
test_atm_withdrawal.py::test_mask_card_no_crash FAILED
test_atm_withdrawal.py::test_average_withdrawal_no_crash FAILED

================================================================================== FAILURES ==================================================================================
_____________________________________________________________________________ test_notes_needed_no_crash _____________________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 5, in test_notes_needed_no_crash
  |     def test_notes_needed_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 6, in test_notes_needed_no_crash
    |     notes_needed(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 2, in notes_needed
    |     return int(amount.replace(',', '')) // 100
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Falsifying test case: test_notes_needed_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 6, in test_notes_needed_no_crash
    |     notes_needed(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 2, in notes_needed
    |     return int(amount.replace(',', '')) // 100
    |                ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'replace'
    | Falsifying test case: test_notes_needed_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

```
_____ test_mask_card_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 9, in test_mask_card_no_crash
  |     def test_mask_card_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 10, in test_mask_card_no_crash
    |     mask_card(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 6, in mask_card
    |     return "XXXX XXXX XXXX " + groups[3]
    |                                ~~~~~~^^^
    | IndexError: list index out of range
    | Failing test case: test_mask_card_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 10, in test_mask_card_no_crash
    |     mask_card(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 5, in mask_card
    |     groups = card_number.split()
    |              ^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_mask_card_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____ test_average_withdrawal_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 13, in test_average_withdrawal_no_crash
  |     def test_average_withdrawal_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 14, in test_average_withdrawal_no_crash
    |     average_withdrawal(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 9, in average_withdrawal
    |     amounts = [int(a) for a in amounts_csv.split(',')]
    |                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 9, in <listcomp>
    |     amounts = [int(a) for a in amounts_csv.split(',')]
    |                ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_average_withdrawal_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\test_atm_withdrawal.py", line 14, in test_average_withdrawal_no_crash
    |     average_withdrawal(s)
    |   File "D:\5TH SEM\SE\SE_LAB\Lab 7\atm_withdrawal.py", line 9, in average_withdrawal
    |     amounts = [int(a) for a in amounts_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_average_withdrawal_no_crash(
    |     s=None,
    | )
    +------------------------------------
```

```
=========================== short test summary info ===========================
FAILED test_atm_withdrawal.py::test_notes_needed_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_atm_withdrawal.py::test_mask_card_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_atm_withdrawal.py::test_average_withdrawal_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 1.14s ==============================

D:\5TH SEM\SE\SE_LAB\Lab 7>
```

```
D:\5TH SEM\SE\SE_LAB\Lab 7>python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): 50000
Number of 100 Notes: 500

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 8888 8888 8888 8888
Masked Card: XXXX XXXX XXXX 8888

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 400,5000,48,55
Average Withdrawal: 1375

D:\5TH SEM\SE\SE_LAB\Lab 7>
```

AFTER FIXING THE BUG

```
D:\5TH SEM\SE\SE_LAB\Lab 7>python -m pytest -v
============================= test session starts =============================
platform win32 -- Python 3.11.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Dhanya KM\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: D:\5TH SEM\SE\SE_LAB\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 3 items

test_atm_withdrawal.py::test_notes_needed_no_crash PASSED
test_atm_withdrawal.py::test_mask_card_no_crash PASSED
test_atm_withdrawal.py::test_average_withdrawal_no_crash PASSED

============================== 3 passed in 0.97s ==============================

D:\5TH SEM\SE\SE_LAB\Lab 7>
```

```
D:\5TH SEM\SE\SE_LAB\Lab 7>python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): 15000
Number of 100 Notes: 150

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 2222 2222 2222 2222
Masked Card: XXXX XXXX XXXX 2222

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 500,400,58,29
Average Withdrawal: 246

D:\5TH SEM\SE\SE_LAB\Lab 7>
```